

Effect of carbon fibre content on friction and wear properties of carbon fibre reinforced PA6 composites

W. Z. Nie¹ and J. Li^{*2}

The tribological performance of polyamide 6 (PA6) and carbon fibre reinforced PA6 (CF/PA6) under dry sliding condition was examined. Different contents of CFs were employed as reinforcement. All filled and unfilled PA6 composites were tested against CGr15 ball, and representative testing was performed. The effects of CF content on the friction and wear properties of the composites were investigated. The worn morphologies of neat PA6 and its composites were examined by SEM, and the wear mechanisms were discussed. Moreover, all filled PA6 have superior tribological characteristics to unfilled PA6. The optimum wear reduction was obtained when the CF content is 20 vol.-%.

Keywords: Tribological, Polyamide 6, SEM, Wear reduction

Introduction

Aromatic polyamides (aramids) exhibit a number of interesting properties, such as high thermal stability, chemical resistance, low flammability and excellent mechanical properties. Wholly aromatic polyamides are a potential thermoplastic material for different applications because of its attractive combination of chemical, physical and mechanical properties. In recent years, considerable research has been devoted to the preparation of polyamide 6 (PA6) reinforced composites.^{1,2} Nevertheless, aromatic polyamides of the class are generally difficult to fabricate with resin matrix because of the immiscibility of these systems. To further improve the mechanical and tribological performance of polyamide based friction material, one of the most efficient methods is to add various kinds of fibres into the matrix as reinforcement. The addition of fibrous reinforcement plays a major role in maintaining strength, stiffness, thermal stability and frictional properties of the composite material.³

Fillers, in the form of particulates and fibres, are often added to polymeric materials to improve their stiffness and strength. This second phase filler material will influence the wear resistance of the composite material. There are many references that illustrate the influence of fillers and fibre reinforcement on the abrasive wear resistance of polymeric composites.^{4,5} Fibre reinforcements, e.g. carbon, glass and aramid fibres, are the main candidates and have been widely employed. Many investigations have shown that the incorporation of fibre reinforcement improved the wear resistance and reduced the coefficient of friction.^{6–11}

Attempts to understand the modifications in the tribological behaviour of the polymers with the addition of fillers or fibre reinforcements have been made by many researchers.^{9,12} Except for a few exceptions,¹³ the wear resistance was improved with lower fibre content generally, e.g. glass fibre or carbon fibre (CF) filled polyethylene naphthalate, polyether ether ketone, polyamide, etc.^{14,15} This was attributed to a reduced ability of ploughing, tearing and other non-adhesive components of wear by the fibres, provided that good interfacial adhesion between the matrix and the fibre reinforcement existed.¹³ Carbon fibre reinforcement dominates in high performance applications due to its outstanding mechanical properties combined with low weight.

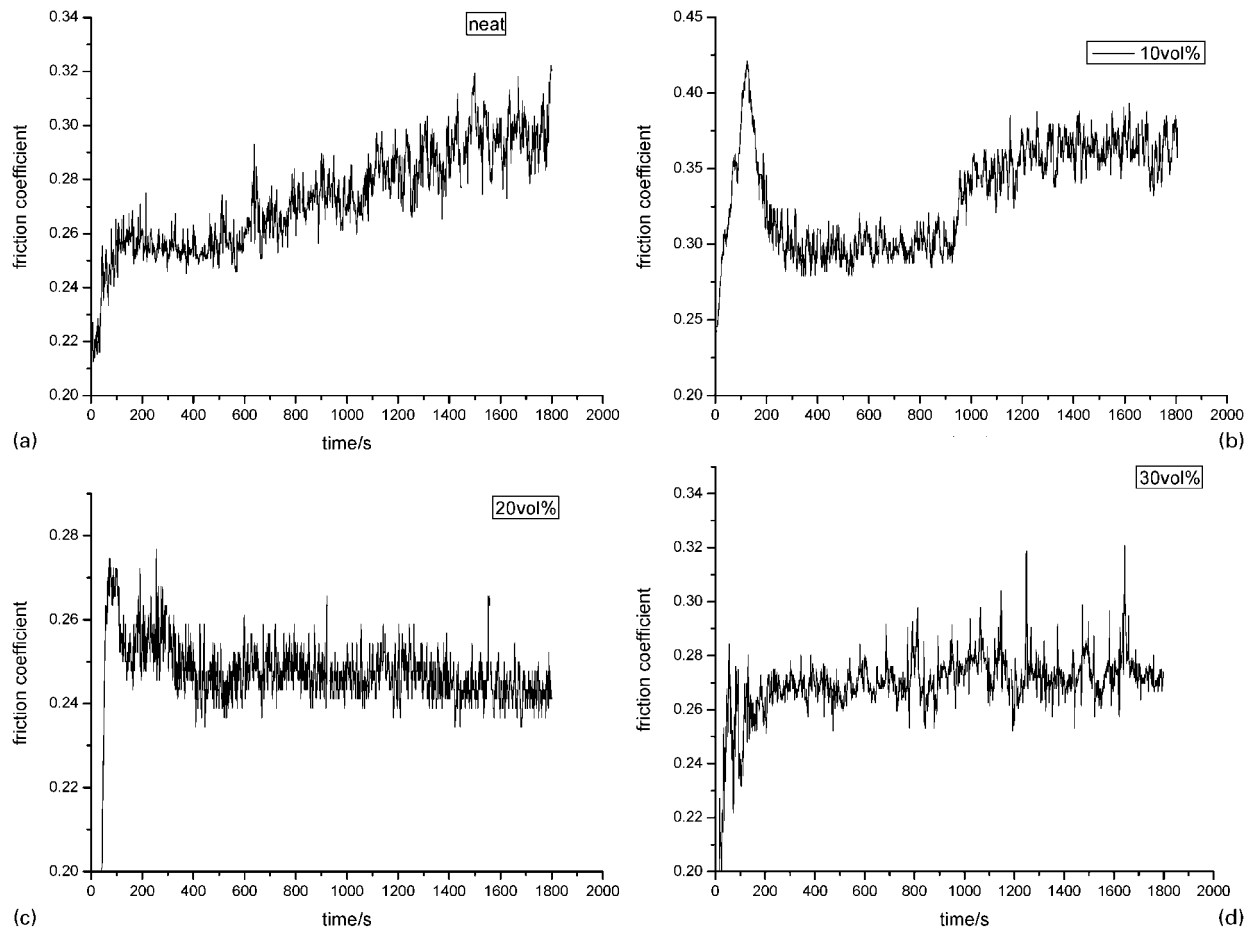
Under controlled testing, a given phase shows a specific wear mode and wear rate, which is determined by its individual properties. Consequently, when various phases are combined to form a multiphase material, it is expected that the overall performance will be a function of the respective contribution of each phase.¹⁶ Nevertheless, the influence of the structure of composites on abrasive wear is a complex function of the properties and interactions of the matrix, the reinforcing constituent and the interface between them, and experimentally, it is found that fillers can either enhance or degrade the wear resistance of polymeric composites.¹⁷

The research on the friction and wear of thermoplastic PA6 and its composite is few. The purpose of the present paper was to clarify the tribological behaviour of neat PA6 and CF reinforced PA6 composites sliding against CGr15 ball under dry sliding condition, so as to provide some practical guidance for the use of this kind of composites under dry sliding condition. The effects of filler content on the tribological properties were also comparatively discussed, and the wear mechanisms of the composites were discussed based on the SEM examination of the worn surfaces.

¹School of Mechanical and Automation Engineering, Shanghai Institute of Technology, Shanghai 200235, China

²School of Mechanical and Electronic Engineering, Shanghai Second Polytechnic University, Shanghai 201209, China

*Corresponding author, email lijian2008d@sina.com



a neat PA6; b 10 vol.-% CF/PA6 composite; c 20 vol.-% CF/PA6 composite; d 30 vol.-% CF/PA6 composite

1 Friction coefficient versus sliding time at 9 N normal load and 4 Hz reciprocating sliding frequency

Experimental

Materials and specimens

For the present investigation, the matrix was PA6 supplied by YueYang Juli Engineering Plastic Co., Hunan with the following specified properties: tensile strength, 85 MPa; flexural strength, 115 MPa; and density, 1150 kg m⁻³. The reinforcements were polyacrylonitrile based high strength CFs. The properties of CFs are given in Table 1.

Preparation process

The composite specimens were prepared by screw inline type injection moulding machine, and the component of

composite includes 10, 20 and 30 vol.-% CFs respectively. The injection moulding technique is according to Table 2.

Friction and wear tests

Friction and wear tests were done using a ball on block reciprocating UMT-2MT tribometer at room temperature with a relative humidity of 30–45%. The specimens were polished using a fine grade SiC emery paper and cleaned ultrasonically with acetone and dried before testing. The reciprocating friction stroke was 5 mm, and tests were conducted at a normal spring driven load. Five tests were conducted under each test condition, and the average values of measured friction coefficient and wear volume were used for further analysis.

Results and discussion

Measured friction coefficient

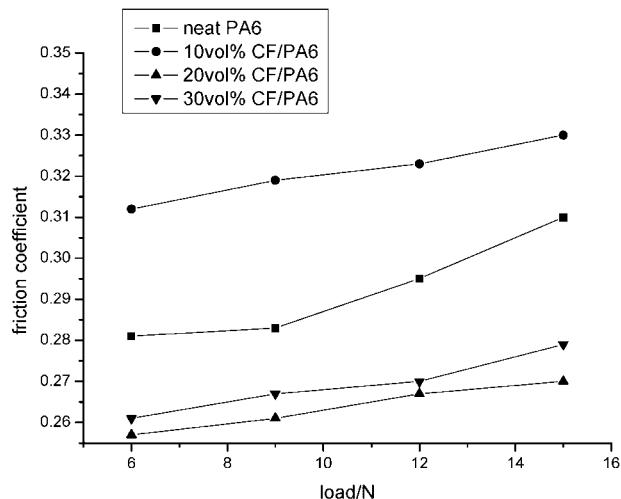
Presented in Fig. 1 are the typical evolutions of the friction coefficient of neat PA6 and CF/PA6 composite as a function of time at the reciprocating sliding frequency of 4 Hz under 9 N applied load. In this study,

Table 1 Properties of carbon fibres

Carbon fibre	
Density, kg m ⁻³	1740
Linear density, mg m ⁻¹	56
Tensile strength, MPa	2000
Linear resistance, Ω m ⁻¹	650
Diameter, µm	7

Table 2 Injection moulding technique

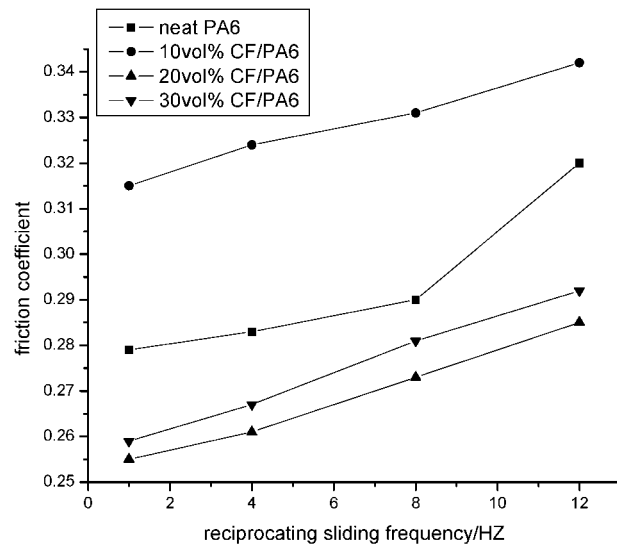
Charging barrel, °C						
Discharge jet	Leading portion	Intermediate section	Posterior segment	Mould temperature, °C	Injection pressure, MPa	Moulding cyclic, s
365	360	355	340	120–150	150–200	20–50



2 Variation of friction coefficient with load (reciprocating sliding frequency: 4 Hz)

the mean value of friction coefficients of the last 1600 s sliding time was taken for evaluating the friction coefficient of the PA6 and CF filled PA6 composite. It can be seen that the friction coefficient of neat PA6 increases quickly during the whole process, while that of 20 and 30 vol.-% CF/PA6 significantly follows a parabolic pattern: the friction coefficients increase with respect to time and arrive at a relatively stable value at last. For 10 vol.-% CF/PA6, the friction coefficient increases during early testing and then decreases to a steady state, thereafter suddenly increases. After that, it stays at a constant level for the remaining period of the test.

The variations of friction coefficient of neat PA6 and CF/PA6 composite are shown in Figs. 2 and 3. It is seen in Fig. 2 that friction coefficients of all filled PA6 composites and neat PA6 increase as the load increases from 6 to 15 N under the same reciprocating sliding frequency 4 Hz, and the friction coefficient decreases as the reciprocating sliding frequency increases from 1 to 12 Hz under the same load 9 N (Fig. 3). The composite exhibits better friction behaviour since the reinforcing CFs can reduce effectively the adhesion force and the plough. Moreover, 20 vol.-% CF reinforced PA6 composite exhibits the lowest friction coefficient. The addition of the CFs strengthened the combination of the interface between the fibres and the PA6 matrix and increased the elastic modulus of the PA6 composites. This would be the reason why the friction coefficient of the CF/PA6 was reduced. As a hard phase in the soft polymer matrix, CFs can reduce the true contact area with the counterbody under certain load.¹⁸ As a result, it exhibits an important influence on reducing the plough and the adhesion between the relative sliding parts. The CFs influence the tribological behaviour of CF/PA6 in two aspects: first, it could lead to fractures in the interface of the two constituents; moreover, it could reduce the plough and the adhesion between the two sliding bodies. These two roles simultaneously influence the friction and wear behaviour according to the sliding conditions. Interfacial fractures will appear on worn surface of CF/PA6. Filled with lower content CFs, the composite exhibits higher friction coefficient compared with pure PA6, so the higher friction coefficients in these cases appear to be derived from the activation of

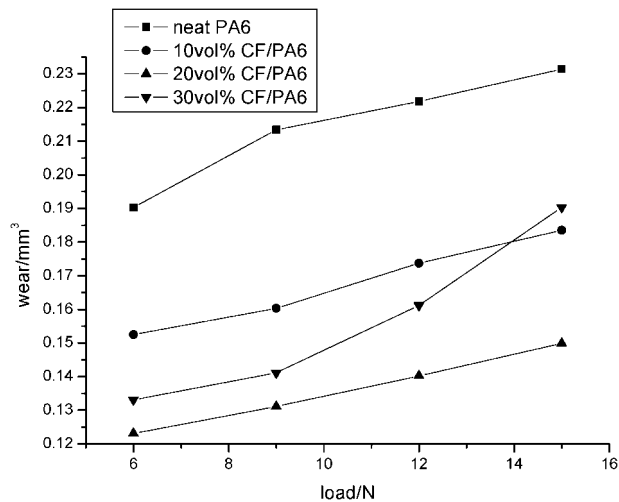


3 Variation of friction coefficient with reciprocating sliding frequency (load: 9 N)

fracture in the interface of reinforcing fibres and PA6 matrix as interfacial energy dissipation mechanism during the sliding process.¹⁹ While for PA6 composite filled with higher CF content, the lower coefficients of the friction compared with pure PA6 may result from the role that the CF played, as CF is a graphitised carbon with the hexagonal planes of its crystals aligned perpendicular to the fibre axis. The lubricating function of graphitised carbon is thought to be responsible for the reduction in friction coefficient, as its composites slide against steel.^{20–22} It is a well documented fact that the CF can decompose into graphite crystals, which have very good lubricating properties.²³ The graphite debris forms a thin lubricating film on the counterface, thereby reducing the abrasion process drastically. This resulted in low friction coefficient and low wear of the composite compared with neat PA6. With increase in reciprocating sliding frequency, the extent of fibre damage and compaction of graphite crystals increased, resulting in more reduction in friction coefficient. A few fibres are microcracked, some are pulverised and some are broken into long pieces. The multiple layers of granular wear debris of the CFs accumulated near the edge. Such compacted particles of CFs are responsible for very low friction coefficient due to their inherent lubricating action, while the increase in load will increase the temperature of composite more remarkably; thus, the adhesion effect will increase the friction coefficient.

Wear of neat PA6 and CF/PA6 composites

Figure 4 shows the volumetric wear of unfilled PA6 and the CF/PA6 composite as a function of the load applied. In general, the volumetric wear for unfilled and all filled PA6 composites increased with increasing applied load, but they exhibit different relationships between volumetric wear and load. The volumetric wear of the neat PA6 and PA6 composites filled with 10 or 20 vol.-% of filler CFs showed little increase with load, while that of the PA6 composites filled with more filler fibres (30 vol.-%) showed a sudden increase between loads of 9 and 15 N. Moreover, it can be seen from Fig. 4 that the addition of CFs can improve the wear resistance of the PA6 composites, and 20 vol.-% CF reinforced PA6



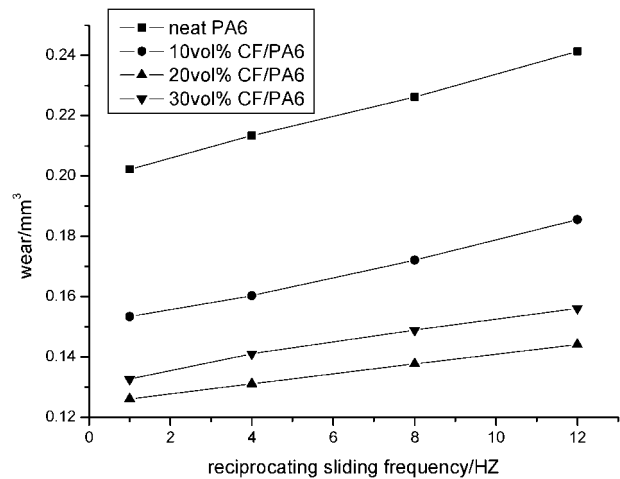
4 Variation of friction coefficient with load (reciprocating sliding frequency: 4 Hz)

composite exhibits the lowest volumetric wear. The unfilled PA6 composite showed, in general, the highest volumetric wear of all the specimens tested under the same sliding condition. In addition, the high percentage (30 vol.-%) of fillers in composite degraded the wear resistance of the CF/PA6 because the fillers themselves caused stress concentrations in the matrix. Furthermore, the detachment of fillers causes the adjacent matrix to be poorly supported and hence is subjected to greater stress and thus more susceptible to fracture.

Figure 5 also shows clearly that, in most test conditions, 20 vol.-% CF reinforced PA6 composite exhibits the lowest volumetric wear, while neat PA6 shows the highest. The increase in reciprocating sliding frequency provokes evident varieties of volumetric wear.

The dependence of the volumetric wear on reciprocating sliding frequency can be attributed to the friction induced heat effects. Most energy dissipated during sliding is transformed into heat, and a high temperature gradient develops in the normal direction to the surface.²⁴ As a result of the low thermal conductivity of PA6, frictional heat being generated during sliding surely provoked an increase in the contact temperature, and the increase in reciprocating sliding frequency can be quantitatively correlated to the increase in sliding temperature.²⁵ Thus, the deterioration of mechanical properties such as hardness and shearing strength occurs as a result of interfacial temperature rising caused by friction induced heat. Accordingly, the applied load can lead to more severe deformation of the composite under higher reciprocating sliding frequency.

Some frictional work was required to drag or to dig the CFs off the matrix since the CFs were limited by the PA6 matrix. These help to reduce material adhesion to the disc surface and bulk temperature of the wear surface. Therefore, the addition of some certain CFs can improve the wear resistance of the PA6 composites. As the wear reducing mechanism of reinforced fibres, Hanmin *et al.*²⁶ and Friedrich *et al.*²⁷ proposed that the fibres support a great portion of the applied load to reduce the direct interaction between polymer and counterface. However, when the CF content was over a certain one, the wear resistance was, in turn, reduced because the fibres would be easily cracked or even dragged due to the lower continuity of the PA6 matrix



5 Variation of friction coefficient with reciprocating sliding frequency (load: 9 N)

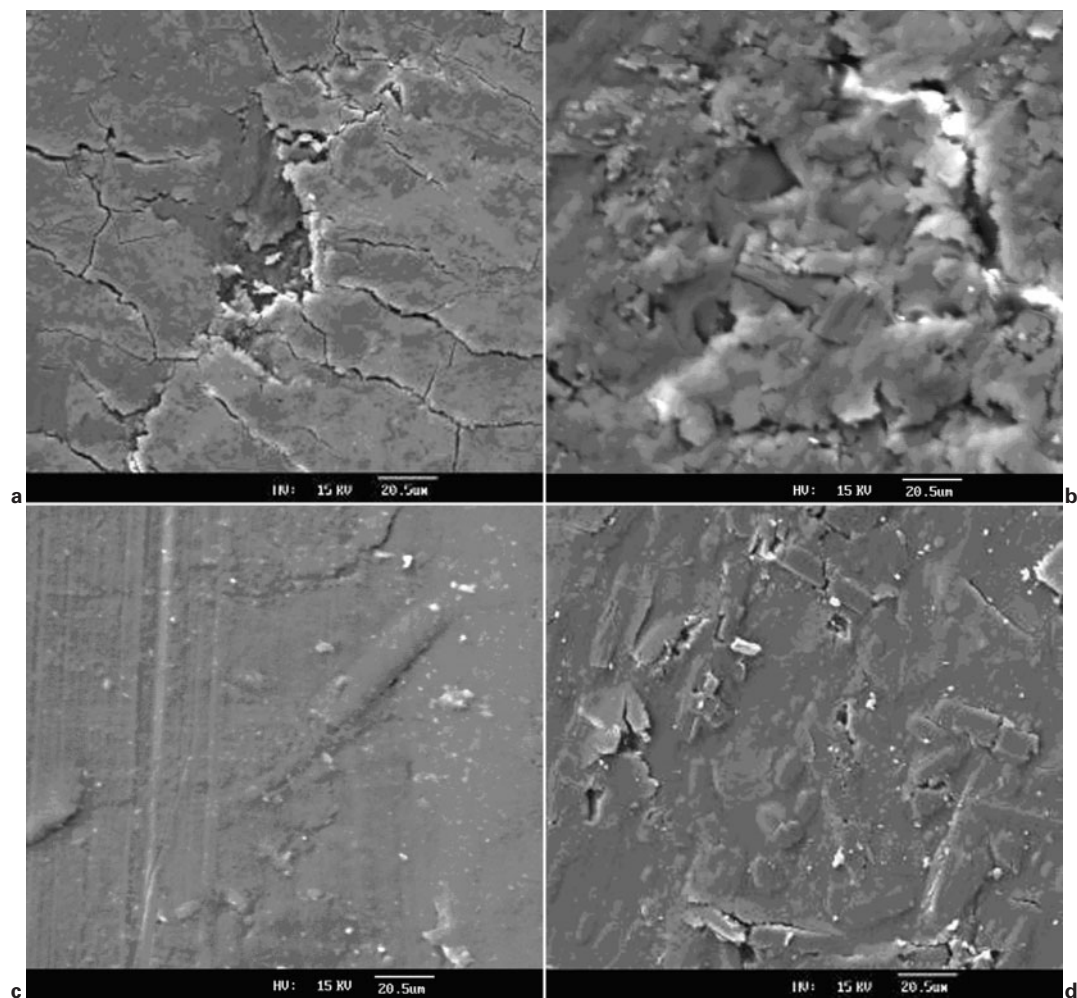
and poor harmonisation of deformation of the fibres with the matrix under the alternative interaction of the asperities of the counterface during sliding. Moreover, stress concentration easily occurred at the interface due to inhomogeneous plastic deformation between reinforcement and matrix and provides preferential sites for stress concentration.^{28,29} The composite that contained a higher fibre fraction may produce more abrasion debris, which contributes negatively to the wear resistance. The higher the CF content, the more the stress concentration sites will be.

Scanning electron microscopy studies on worn surfaces

The worn surfaces of the neat PA6 and CF/PA6 composite under the same load and reciprocating sliding frequency are shown in Fig. 6. In the case of unfilled PA6 (Fig. 6a), microcracks and peeling of PA6 were observed. Microcracking of the matrix material was the main wear mechanism operative. The matrix material exhibited very poor wear resistance in wear tests as it was removed. The microcracks would be originated and extended through the contacting surface layer of the PA6 composites in sliding, since there exists the tensile stress on the surface layer behind the contacting asperities and the compressive stress on the surface layer before the contacting asperities, which resulted in a strain fatigue wear. A topography like this is conducive to abrasion of the soft polymer material, and so the volumetric wear would be high.

Some debris was attached in the form of flakes. This feature suggests that fatigue wear is the main wear mechanism of CF/PA6 composite. The worn surface of the 10 vol.-% CF/PA6 sample is shown in Fig. 6b. Microcracks were observed at the surface either at the fibre-matrix boundary or at weak spots in the matrix and eventually led to delamination of the matrix material. Low content of CF cannot support the load from the counterface sufficiently. This means that the matrix far away from the CF has the same wear mechanism with neat PA6.

Figure 6d shows a large crack in the matrix, normal to the sliding direction, which formed after neighbouring fillers had been detached. This would give rise to a high volume loss of matrix. Microcracking and subsequent spalling of material are important wear modes for high



a neat PA6; b 10 vol.-% CF/PA6 composite; c 20 vol.-% CF/PA6 composite; d 30 vol.-% CF/PA6 composite
6 Images (SEM) of worn surface of neat PA6 and CF/PA6 composites at load of 9 N and reciprocating sliding frequency of 4 Hz

content filled polymeric materials. The fibre tips were clearly observed, protruding out of the polymer surface. Evidently, this is so because CF has much better mechanical properties and wear resistance compared to matrix material. Probably, a crack follows the fibre–matrix interface and passes between the fibres at their closest distance. The crack propagates under the original surface matrix layer and causes fragments of the matrix to be broken off, leaving the fibres bare. The driving force for the crack comes from the friction forces being applied on the matrix surface. Where the fibres are close to each other, the matrix between the fibres are often fragmented and broken off when the crack propagates along the fibre surface. In addition, the fibres separated from the surface layer at the sliding contact interface of the PA6 composites resulted in the occurrence of the three body abrasion. Plastic deformation occurred under sliding load together with some irregular shape debris. Figure 6d also shows that large filler fibres were fractured into fragments, and many small filler particles were detached from the matrix material, leaving cavities in the matrix. These cavities were themselves stress concentrations and resulted in more cracks in the matrix and a higher volumetric wear. Large sections of matrix were removed causing even larger filler particles to be detached from the matrix. Thus, the fillers do not support the load resulting in a sudden increase in the

volumetric wear. In conclusion, wear of PA6 composites is influenced by the properties of the filler, matrix and interface by the relative hardness of the filler to that of the abrasive grit or counterface, by the content, distribution of filler and by the abrasiveness of filler against the matrix. In addition, the cavity shown in the matrix is the result of a filler particle detaching from the matrix due to loss of matrix around it and poor adhesion between the filler and matrix.

While for the 20 vol.-% CF/PA6 composite (Fig. 6c), the worn surface is relatively smooth and the peeling of matrix and CFs are constrained due to fact that the CFs effectively support the load from the counterface. That is to say that the CF and matrix combine well.

Conclusions

1. The incorporation of CF into PA6 can either increase or decrease friction coefficient and reduce volumetric wear of the materials in sliding against stainless steel under dry sliding condition. The optimum wear resistance property was obtained at the CF content of 20 vol.-%.

2. The friction coefficient of neat PA6 and CF/PA6 composites increases with the increase in the load and decreases with the increase in the sliding frequency, while the volumetric wear of neat PA6 and CF/PA6

composites increases with the increase in load and reciprocating sliding frequency.

3. The adherence and plastic deformation are primary wear mechanisms for the neat PA6 under dry sliding. When incorporated with CF, the adherence and plastic deformation are greatly reduced.

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